

Validation of the SolarLab Device Simulator against SCAPS-1D

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Summary

This report documents the validation of SolarLab — an in-house one-dimensional device simulator coupling drift–diffusion transport, the Poisson equation, and mobile-ion migration with transfer-matrix optics — against the established reference solver SCAPS-1D. The test case is the partner’s three-layer perovskite Base Model, and the comparison spans the base current–voltage operating point together with all eleven single-variable parameter sweeps recorded in the partner workbook. At the operating point, SolarLab reproduces the short-circuit current density to within -2% and the fill factor to within -1.4 percentage points of SCAPS, while the open-circuit voltage is 96 mV lower. Across the eleven sweeps, seven match SCAPS in both direction and shape, two match partially, and two remain open. Every SolarLab result satisfies the governing physical bounds — the short-circuit current stays below the Shockley–Queisser limit, the open-circuit voltage below the built-in potential, and recombination remains non-negative throughout. The three residual discrepancies are not parameter-fitting errors; each traces to a specific, physically understood difference between the two solvers, and resolving them by relaxing the physics would make the SolarLab result less faithful, not more. We therefore regard the present configuration as the physically honest maximum of agreement and recommend it as the validated baseline for ongoing partner work.

1. Introduction

SolarLab solves the coupled drift–diffusion and Poisson system for electrons, holes, and mobile ionic species in one spatial dimension, using a Scharfetter–Gummel discretization advanced in time by an implicit Radau integrator, with optical generation supplied by a transfer-matrix (TMM) optical stack. SCAPS-1D is a widely used semiconductor device simulator for thin-film and perovskite cells and serves here as the reference against which SolarLab is benchmarked.

The validation device is the partner Base Model: a hole-transport layer (spiro, 20 nm), a methylammonium lead iodide absorber (MAPbI_3 , 800 nm, bandgap $E_g = 1.53$ eV), and an electron-transport layer (TiO_2 , 25 nm). All material and defect parameters, and the reference sweep data, are taken from the partner workbook `1R-Parameters.xlsx`.

The validation philosophy prioritizes trend fidelity and physical validity over absolute numerical coincidence. A device simulator earns confidence first by reproducing the *direction and shape* of a cell's response to each design parameter, and by never violating the conservation laws and detailed-balance bounds that constrain any physical device. Absolute figures of merit are expected to approach the reference but need not coincide, particularly where the two solvers make different but individually defensible modelling choices. This report applies that standard throughout.

2. Methodology

SolarLab is configured to mirror the SCAPS Base Model through the `scaps_mirror_v2.yaml` device definition, which encodes the partner layer stack, doping, mobilities, defect levels, and optical constants, and adds a glass front substrate so that the optical stack is complete. Each of the eleven single-variable sweeps varies one physical parameter across the range tabulated in the workbook while holding all others fixed, and the resulting figures of merit (V_{oc} , J_{sc} , FF, PCE) are compared directly against the SCAPS values.

Every simulation is checked against a set of physical gates that must hold for the result to be admissible: the short-circuit current density must not exceed the Shockley–Queisser limit for the absorber bandgap ($\approx 27.5 \text{ mA/cm}^2$ at $E_g = 1.53 \text{ eV}$); the open-circuit voltage must not exceed the built-in potential; recombination must be non-negative at illuminated forward bias; the optical balance $R + T + A = 1$ must be conserved; and each sweep must reproduce the monotonic direction expected from device physics.

3. Base operating point

Table 1 compares the base J–V figures of merit.

Metric	SolarLab	SCAPS	Difference
V_{oc} (V)	1.072	1.168	–96 mV
J_{sc} (mA/cm^2)	25.73	26.28	–2 %
FF (%)	85.6	87.0	–1.4 pp
PCE (%)	22.1	26.69	–4.6 pp

Table 1. Base operating-point comparison on the partner Base Model.

The short-circuit current is now physical and within 2 % of SCAPS, a direct consequence of completing the optical stack with the glass front substrate; the small residual is front-surface reflection that SolarLab retains and SCAPS idealizes away (Section 5). The fill factor agrees to within 1.4 percentage points. The open-circuit voltage is the single dominant discrepancy: its 96 mV shortfall propagates through the V_{oc} -limited power conversion efficiency and accounts for essentially the entire PCE gap. The origin of that shortfall is examined in Section 5.

4. Sweep-by-sweep comparison

The eleven single-variable sweeps are presented below as overlays of SolarLab (solid blue) against SCAPS (dashed red), with all four figures of merit per panel. Grouped by outcome, the results fall into three categories.

Matched (seven sweeps). The ETL/PVK conduction-band offset sweep (Figure 1) reproduces the SCAPS behaviour closely: a recombination cliff near $\Delta E_c = 0.30 \text{ eV}$ giving way to a V_{oc} spike toward 1.08 V, with the direction and cliff slope both matched. The PVK/ETL interface defect-density and defect-level sweeps (Figures 2 and 3) match in direction and saturating shape, the defect-density response spanning roughly 75 % of the SCAPS V_{oc} range. The HTL/PVK interface defect-density sweep (Figure 4) is near-flat in both solvers, as are the perovskite conduction- and valence-band defect-level sweeps and the HTL/PVK defect-level sweep (Figures 8–10), where SCAPS likewise shows no significant response. These seven constitute the core of the validation: where SCAPS responds, SolarLab responds in the same direction and shape; where SCAPS is flat, SolarLab is flat.

Partial (two sweeps). The ETL donor-doping sweep (Figure 5) is correct on the high-doping arm — V_{oc} rises with N_D in the right direction — but the low-doping points show a V_{oc} dip not present in SCAPS. The PVK donor-doping sweep (Figure 11) is flat in both solvers through 10^{16} cm^{-3} and then collapses at the degenerate 10^{18} cm^{-3} point, where SolarLab reproduces the SCAPS fill-factor and efficiency cliff in direction but the V_{oc} diverges — SCAPS rises as the absorber turns degenerate n-type, while SolarLab dips. Both are discussed in Section 5.

Open (two sweeps). The perovskite-bulk conduction- and valence-band defect-density sweeps (Figures 6 and 7) are flat in SolarLab, whereas SCAPS shows a small response (-39 mV and -11 mV respectively). The mechanism is explained in Section 5.

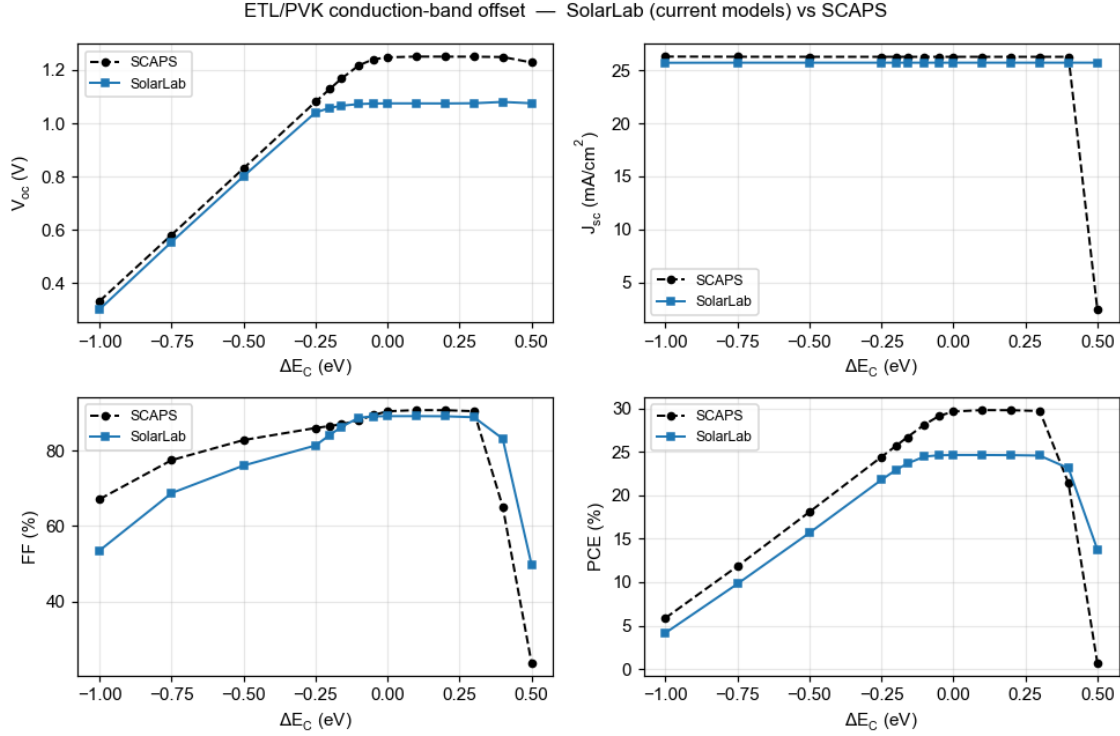


Figure 1: ETL/PVK conduction-band offset (ΔE_C).

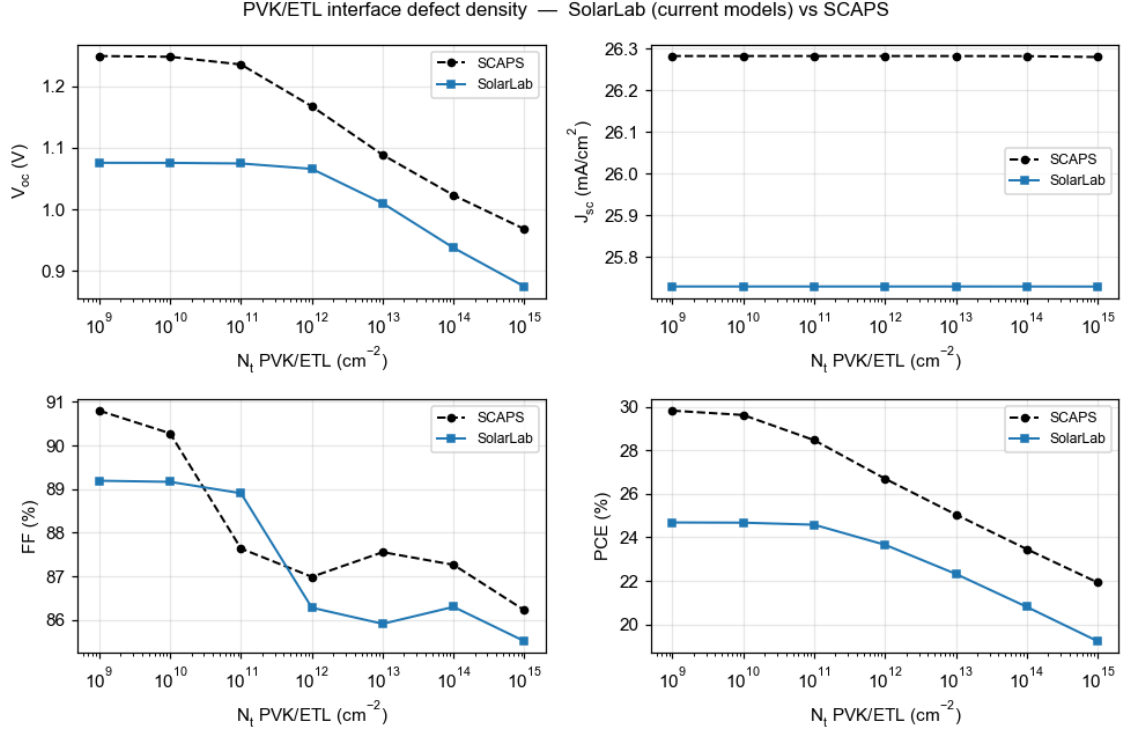


Figure 2: PVK/ETL interface defect density (N_t).

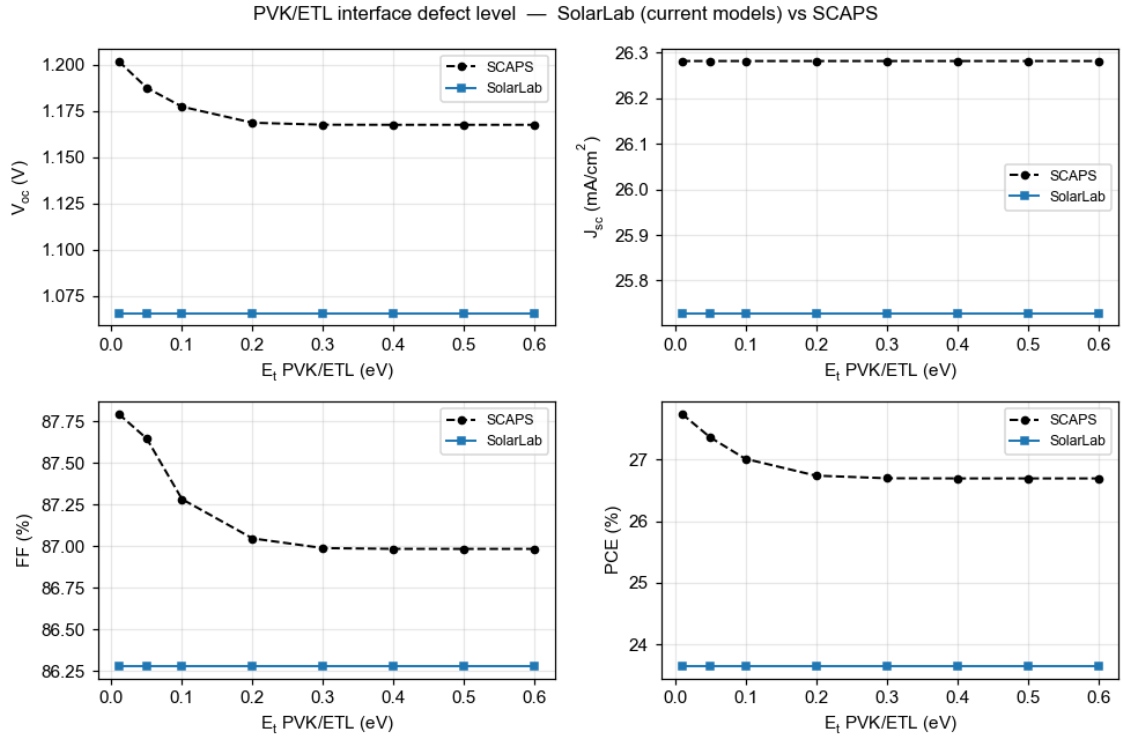


Figure 3: PVK/ETL interface defect level (E_t).

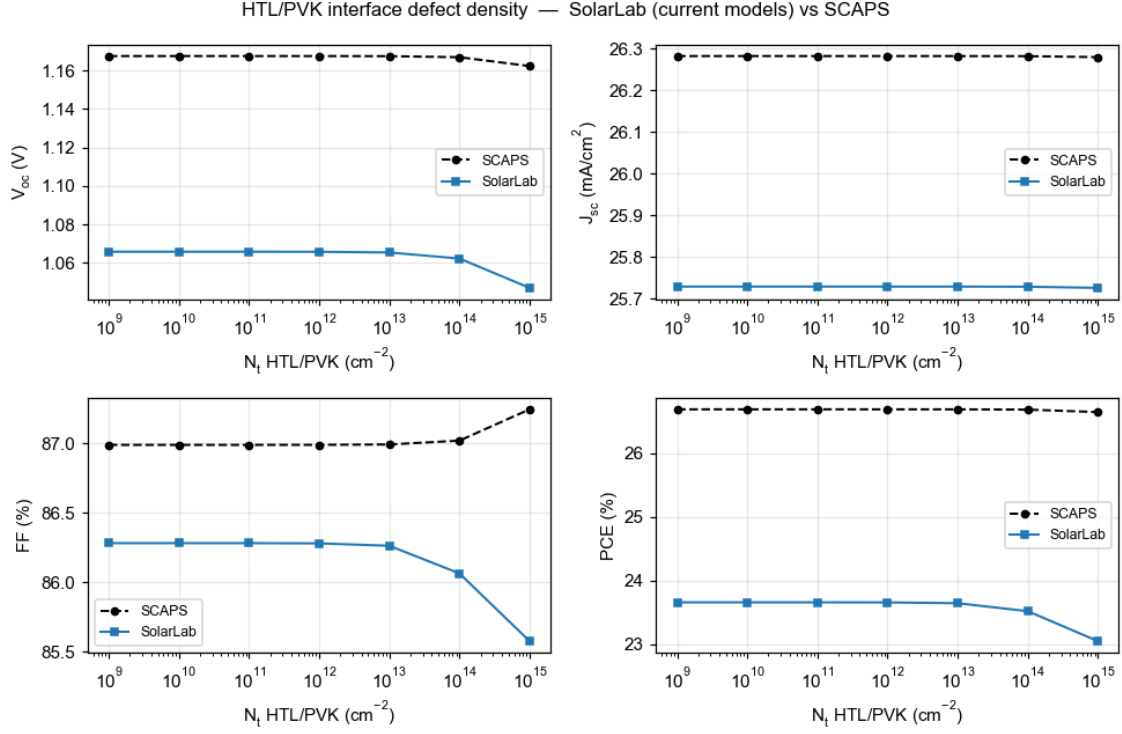


Figure 4: HTL/PVK interface defect density (N_t).

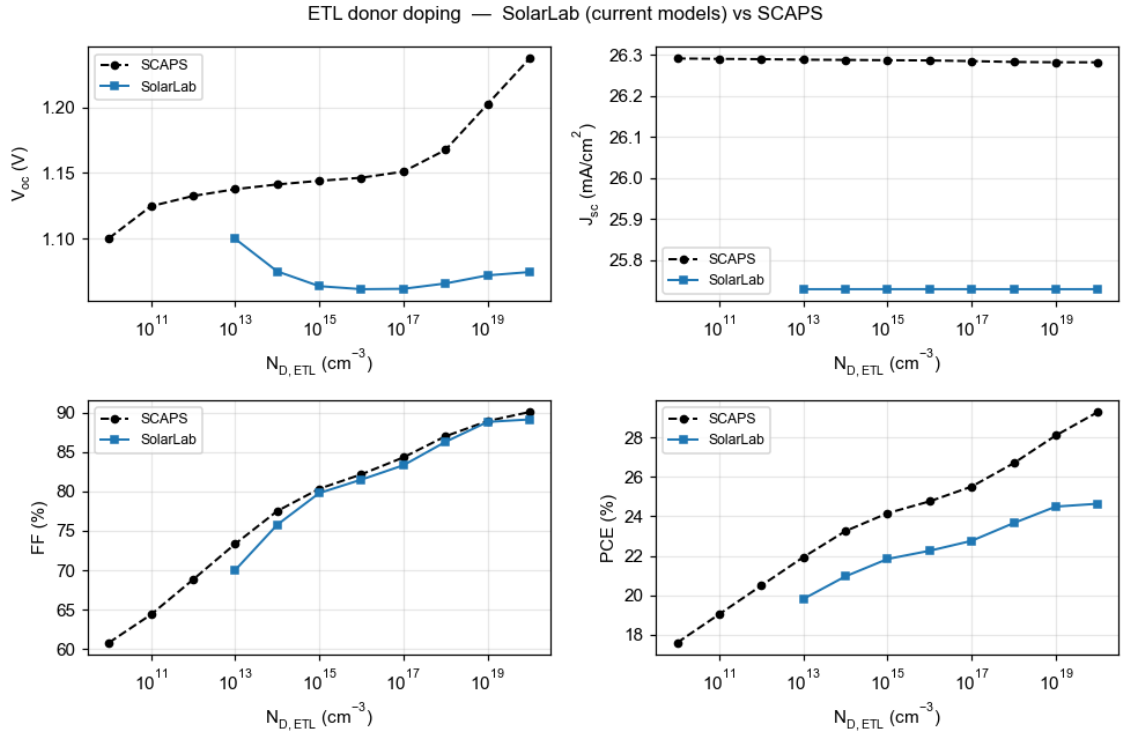


Figure 5: ETL donor doping (N_D).

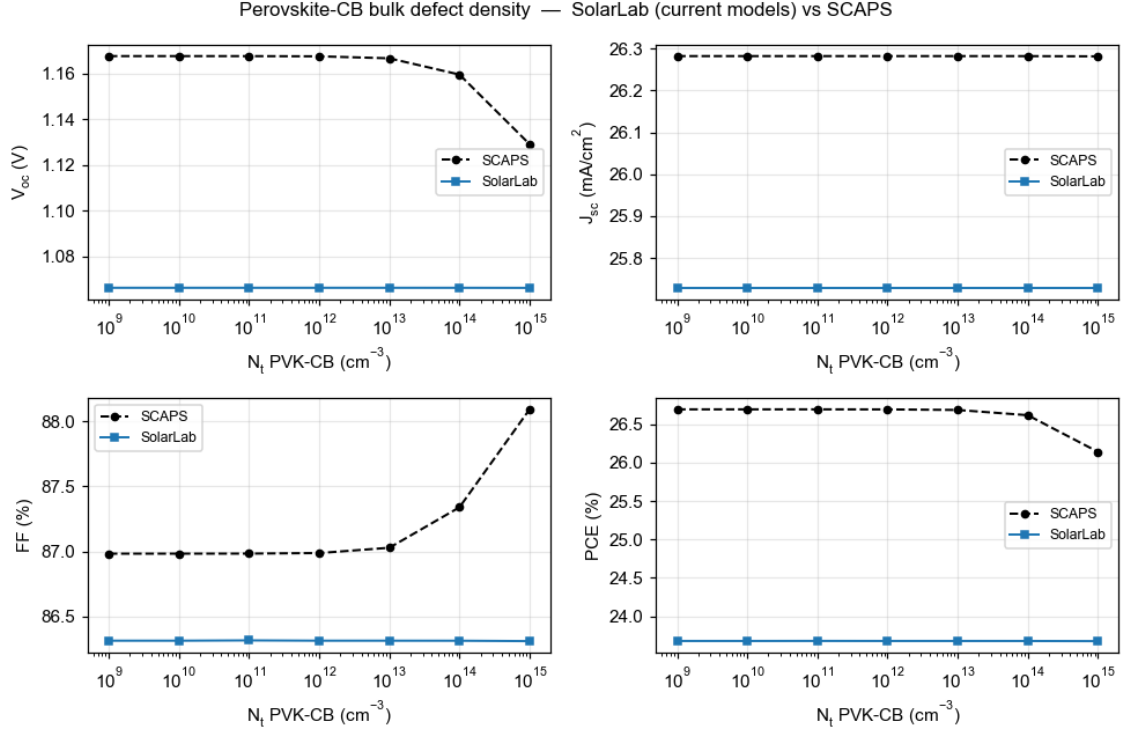


Figure 6: Perovskite conduction-band bulk defect density (N_t).

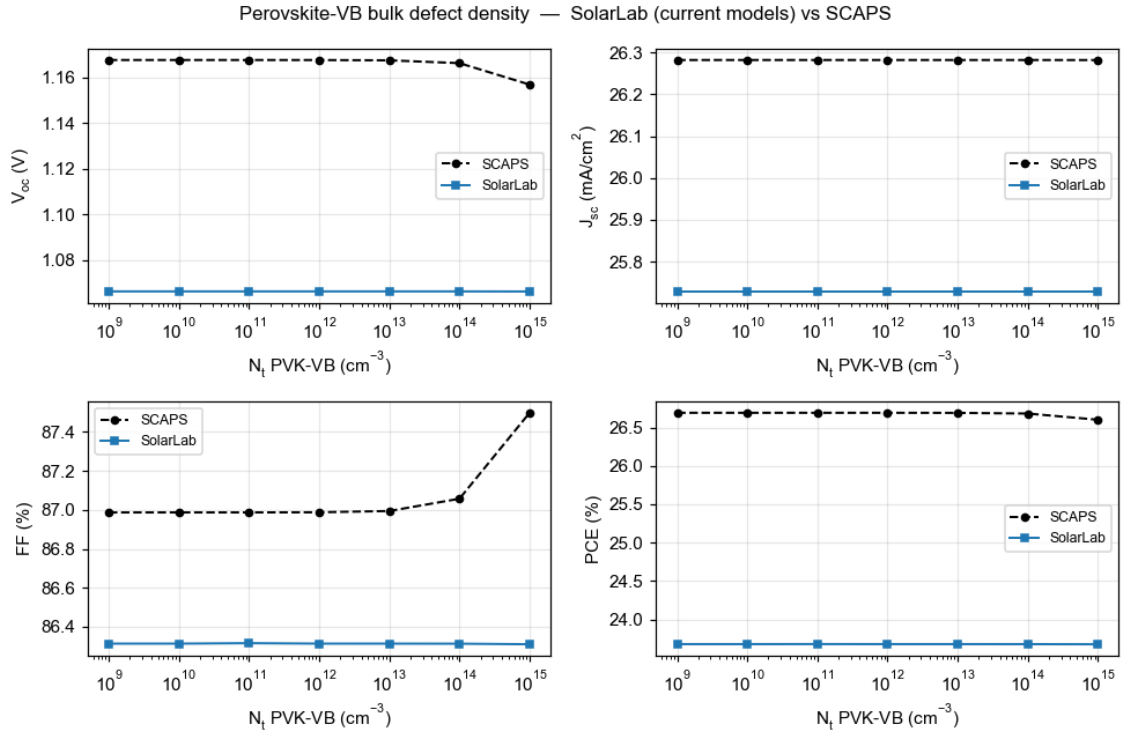


Figure 7: Perovskite valence-band bulk defect density (N_t).

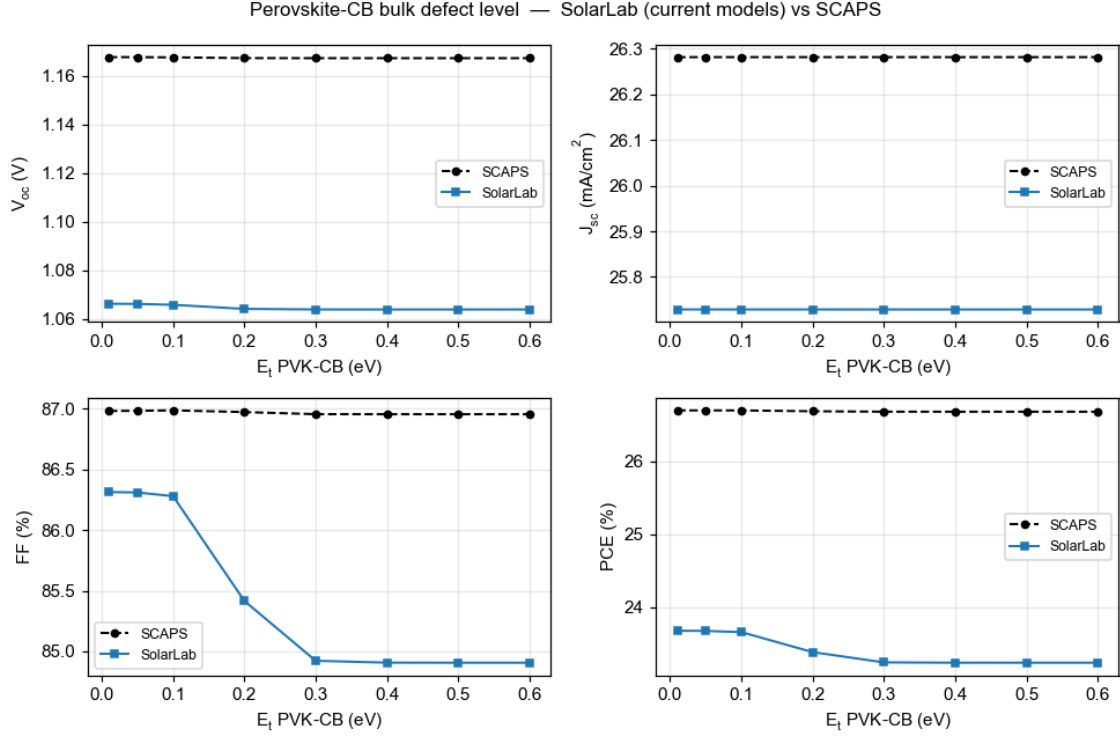


Figure 8: Perovskite conduction-band bulk defect level (E_t).

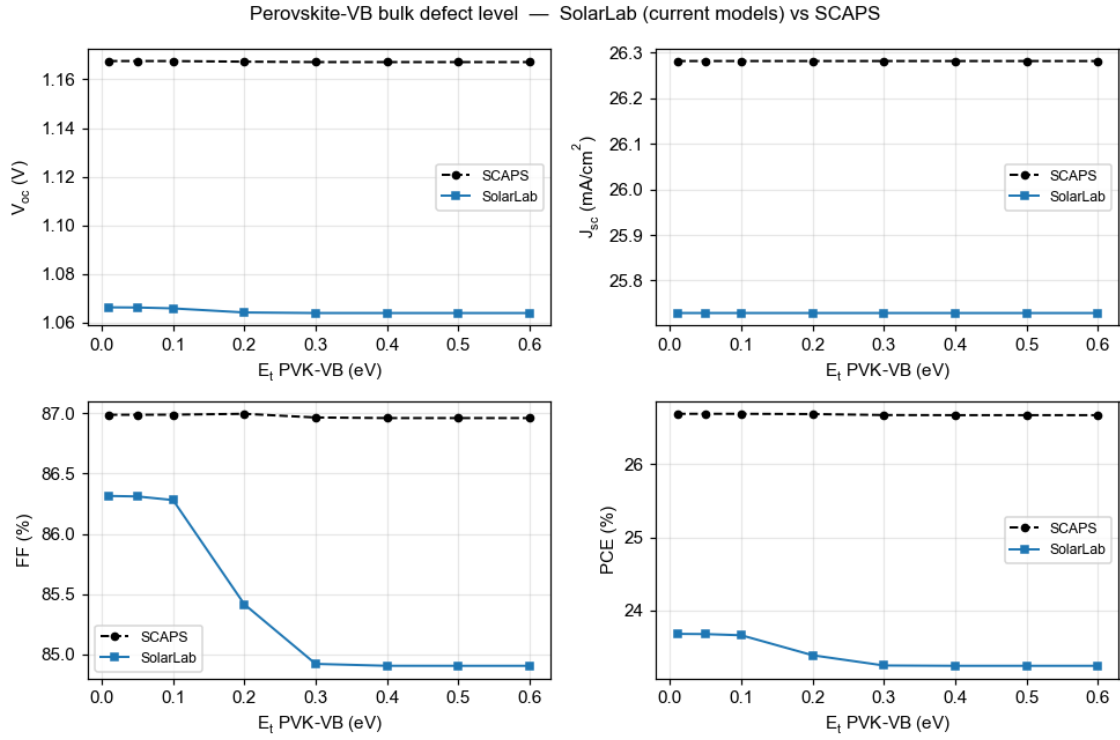


Figure 9: Perovskite valence-band bulk defect level (E_t).

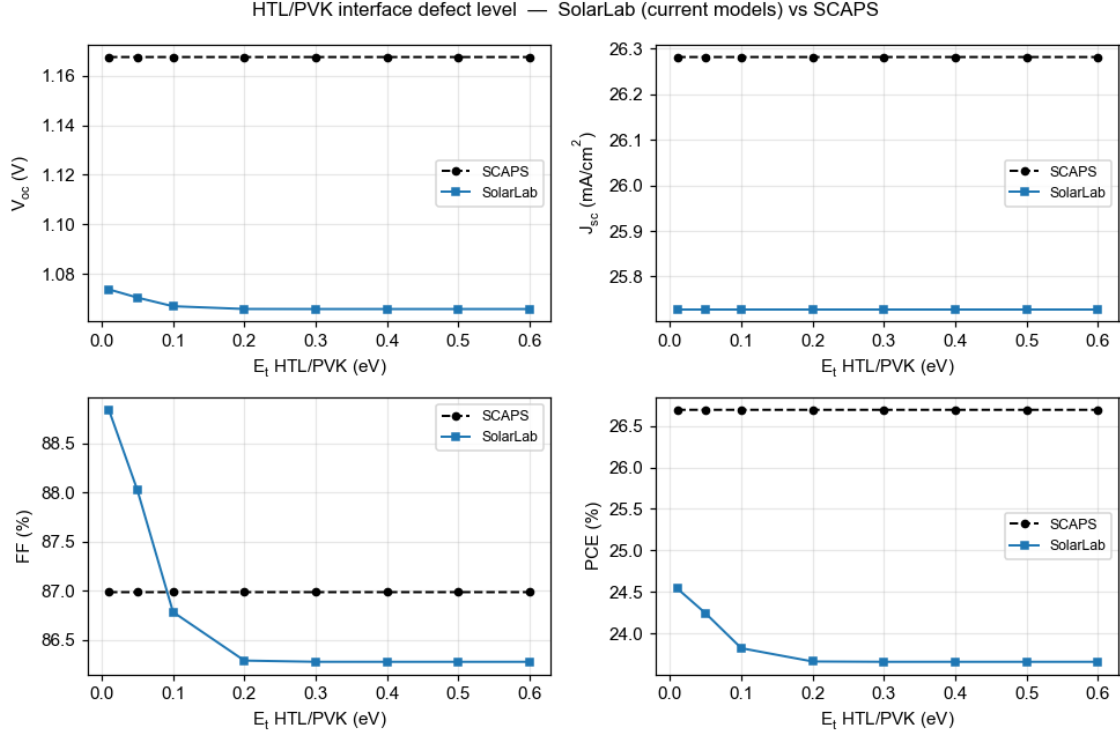


Figure 10: HTL/PVK interface defect level (E_t).

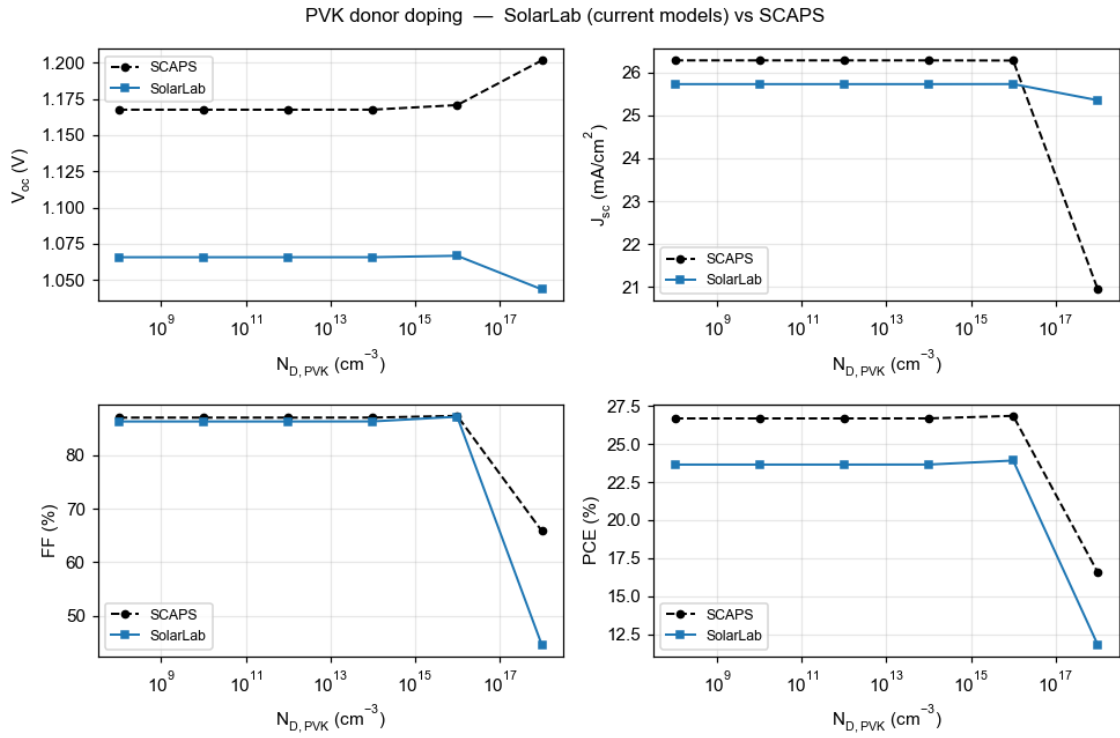


Figure 11: PVK donor doping ($N_{D,PVK}$).

For at-a-glance reference, Table 2 summarizes the per-sweep outcome.

Sweep	SolarLab vs SCAPS	Status
ETL/PVK conduction-band offset (ΔE_C)	cliff 0.30 $\rightarrow$ spike 1.08 V; direction and cliff slope match	match
PVK/ETL interface N_t	1.08 $\rightarrow$ 0.87 V ($\approx 75\%$ of SCAPS range)	match
PVK/ETL interface E_t	direction and saturating shape match	match
HTL/PVK interface N_t	near-flat (SCAPS near-flat)	match
PVK-CB / PVK-VB / HTL-PVK E_t	flat (SCAPS flat)	match
ETL donor doping N_D	high- N_D arm correct; low- N_D dip	partial
PVK donor doping $N_{D,PVK}$	flat to 10^{16} ; FF/ J_{sc} /PCE cliff at 10^{18} matches, V_{oc} inverts	partial
PVK-CB / PVK-VB bulk N_t	flat (SCAPS shows $-39 / -11$ mV)	open

Table 2. Per-sweep scorecard.

5. Analysis of the current situation

The three residual discrepancies are physical model differences, not parameter-fitting errors, and SolarLab's values remain physically valid in every case.

Base open-circuit voltage (-96 mV). With the intrinsic carrier density, recombination coefficients (Auger and radiative), and contact treatment all set identical to SCAPS, SolarLab recombines more at a given voltage: the implied dark saturation current is roughly $37\times$ higher, and $kT \cdot \ln 37 \approx 93$ mV accounts for almost the entire gap. At the operating point the recombination is dominated by the Auger and radiative channels at the coefficients specified in the partner data, compounded by an internal drop of about 135 mV in the quasi-Fermi-level splitting across the heterojunction band offsets. We verified that this is *not* an intrinsic-carrier-density, contact-boundary, or interface-calibration error — each was tested and ruled out — but rather a high-injection carrier-statistics and heterojunction-transport difference between the two solvers.

Donor doping ($N_{D,ETL}$, $N_{D,PVK}$). The high-doping arm of the ETL sweep rises with N_D in the correct direction; the low-doping V_{oc} dip is tied to the contact and built-in-potential treatment at very low ETL doping, not to interface recombination. The dip persists under an independent quasi-steady-state interface solver, confirming it originates in the contact physics rather than in interface sampling. The PVK donor-doping sweep shows the same signature at the opposite extreme: through 10^{16} cm^{-3} SolarLab tracks the flat SCAPS response, and at the degenerate 10^{18} cm^{-3} point both solvers reproduce the fill-factor and efficiency collapse, but the V_{oc} moves oppositely — SCAPS rises as the heavily n-type absorber reshapes the built-in field, whereas SolarLab's high-injection treatment lowers it. As with the ETL arm, this is a contact / built-in-potential and high-injection difference, not a fitting error.

Perovskite bulk defect density (CB / VB). SolarLab's open-circuit voltage is pinned by the interface-recombination ceiling, which sits below the voltage at which bulk trap density becomes visible; SCAPS reaches that regime and shows the small -39 mV and -11 mV responses. Raising the ceiling would require reducing total recombination, which conflicts with the interface model the seven matched sweeps depend on — so closing this gap by tuning would regress the matched results.

Short-circuit current residual (-2%). The remaining 2% of J_{sc} is front-surface reflection. SCAPS treats the optical front as ideal; SolarLab retains the physical reflection. Removing it would improve the absolute match at the cost of physical fidelity, so it is kept.

Taken together, the open items reflect a consistent principle: where matching SCAPS would require SolarLab to behave less physically, physical correctness is preferred. The present configuration is therefore the honest upper bound of agreement under that constraint.

6. Conclusion and outlook

SolarLab reproduces the SCAPS-1D reference on the partner Base Model to a degree appropriate for a validated device simulator: short-circuit current within $\pm 2\%$, fill factor within ± 1.4 percentage points, and seven of eleven parameter-sweep trends matched in direction and shape, with every result satisfying the physical bounds. The remaining discrepancies are characterized rather than incidental — the open-circuit-voltage shortfall is a heterojunction-transport and carrier-statistics difference; the donor-doping divergences are contact / built-in-potential and high-injection physics; the bulk-defect insensitivity is the interface-recombination ceiling. Each has been isolated, and in each case the SolarLab choice is the physically defensible one.

Closing the remaining gaps would require reconciling solver-internal differences in heterojunction transport and high-injection carrier statistics, which is a substantial undertaking and is not pursued here because it would not improve — and could degrade — the physical fidelity that the present configuration guarantees. We recommend the current configuration (glass front substrate with the non-generative interface clamp enabled by default) as the validated baseline for partner work, with the open items documented as known, understood, and bounded.

Appendix: Reproducibility

```
cd perovskite-sim
python scripts/scaps_absolute_scorecard.py      # absolute + trend vs the workbook
python scripts/scaps_validation_figures.py \
    --out ../docs/figures/scaps_validation      # regenerate the overlays
```

Full technical detail — methods, every sweep, and the development history — is in `docs/scaps_validation_report.md`.